

MSTAR PRACTICE WORKSHEET

Lesson 6-5: Write and Evaluate Algebraic Expressions

Grade 6 Mathematics · Standard 6.AT.6a

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which expression represents 'the product of 6 and a number n '?

- ☒ A $6n$
- ☐ B $6 + n$
- ☐ C $n - 6$
- ☐ D $n \div 6$

'Product' means multiplication, so the product of 6 and n is $6n$.

Question 2 SELECTED RESPONSE — CORRECT: A

Which expression represents '9 less than a number y '?

- ☒ A $y - 9$
- ☐ B $9 - y$
- ☐ C $y + 9$
- ☐ D $9y$

'Less than' means subtract from the number, so it's $y - 9$.

Question 3 SELECTED RESPONSE — CORRECT: A

Which expression represents 'the sum of a number m and 15'?

- ☒ A $m + 15$
- ☐ B $m - 15$
- ☐ C $15m$
- ☐ D $m \div 15$

'Sum' means addition, so the sum of m and 15 is $m + 15$.

Question 4 SELECTED RESPONSE — CORRECT: B

Evaluate $4.25s + 23.50$ when $s = 10$.

- ☐ A \$47.75
- ☒ B \$66.00
- ☐ C \$66.75
- ☐ D \$277.50

Substitute $s = 10$: $4.25 \times 10 + 23.50$. Multiply first: 42.50, then add: $42.50 + 23.50 = \$66.00$.

Question 5 SELECTED RESPONSE — CORRECT: A

Which key word tells you to multiply?

- ☒ A Product
- ☐ B Sum
- ☐ C Difference
- ☐ D Quotient

'Product' means multiplication. 'Sum' means addition, 'difference' means subtraction, and 'quotient' means division.

Question 6

SELECTED RESPONSE — CORRECT: C

The expression $8n + 15$ gives the cost of n tickets plus a fee. What is the cost for 6 tickets?

- ☐ A \$29
- ☐ B \$48
- ☒ C \$63
- ☐ D \$138

Substitute $n = 6$: $8 \times 6 + 15 = 48 + 15 = \63 .

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7**TWO-PART QUESTION****Part A — correct answer: A**

A concert ticket costs 8 dollars. The website also charges one 5 dollar service fee for the whole order. Let t represent the number of tickets bought in one order. Which expression gives the total cost in dollars?

- ☒ **A** $8t + 5$
- ☐ **B** $8 + 5t$
- ☐ **C** $8(t + 5)$
- ☐ **D** $13t$

The variable t is the number of tickets. Each ticket costs 8 dollars, so the ticket cost is $8t$. The 5 dollar fee is charged once, so it is a constant added on: $8t + 5$.

- **B:** This charges the fee on every ticket and the ticket price only once. The 8 dollars is per ticket, so it pairs with t .
- **C:** This multiplies the fee by the ticket price, charging 8 dollars on the fee as well. The fee is added once.
- **D:** This charges 13 dollars per ticket, applying the one-time fee to every ticket.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

- ☐ **A** The 5 dollar fee is paid for each ticket, so 5 is the coefficient: $8 + 5t$.
- ☒ **B** The 8 is the coefficient because it is paid for each ticket, and the 5 is a constant because it is paid one time. This gives $8t + 5$, so an order of 6 tickets costs $8 \times 6 + 5 = 53$ dollars.
- ☐ **C** Add the fee to the number of tickets first, then multiply by the ticket price: $8(t + 5)$.
- ☐ **D** Combine 8 and 5 into a single coefficient because both are numbers: $13t$.

A per-item price becomes the coefficient of the variable and a one-time charge becomes a constant. Substituting $t = 6$ into $8t + 5$ gives 53 dollars.

Question 8

SELECT ALL THAT APPLY — CORRECT: A, B, C, D

Select ALL statements that are true about the algebraic expression $7n + 4$.

- ☒ A The variable is n .
- ☒ B The coefficient of the variable term is 7.
- ☒ C The constant is 4.
- ☒ D When $n = 3$, the expression has a value of 25.
- ☐ E When $n = 3$, the expression has a value of 33.
- ☐ F The term $7n$ means 7 plus n .

In $7n + 4$ the variable is n , its coefficient is 7, and 4 is the constant. Substituting $n = 3$ gives $7 \times 3 + 4 = 21 + 4 = 25$. A value of 33 comes from combining 7 and 4 into 11 first, which is not allowed because $7n$ and 4 are not like terms. The term $7n$ means 7 times n .

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

Field day needs one \$4.25 item per student plus a one-time \$23.50 shipping fee. Write the expression, then find the cost for 30 students AND for 60 students. Does doubling the students double the cost? Explain.

Model answer: The expression is $4.25s + 23.50$. For $s = 30$: $4.25 \times 30 + 23.50 = 127.50 + 23.50 = \151.00 . For $s = 60$: $4.25 \times 60 + 23.50 = 255.00 + 23.50 = \278.50 . Doubling the students does NOT double the cost. Twice \$151.00 would be \$302.00, but the actual cost is \$278.50 — a difference of exactly \$23.50. The reason is that only the $4.25s$ term depends on the number of students; the shipping fee is a constant charged once no matter how many students come, so it is not doubled along with everything else.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.